TOAN LE

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EDUCATION

University of California, San Diego **San Diego, CA**
Bachelor of Science, Structural Engineering *Sept 2021 – June 2026*

- **GPA:** 3.16/4.00
- **Relevant Coursework:** Structural Analysis, Steel Design, Finite Element Analysis, Geotechnical Engineering, Earthquake Engineering, Composite Structures, Design of Reinforced Concrete, Design of Civil Structures, Design of Prestressed Concrete

PROJECTS

Design of Civil Structures Capstone I & II **San Diego, CA**
Team Structural Design Project *Dec 2025 – Jun 2026*

- Collaborated on a 3- to 4-student team to design gravity load-carrying systems, buckling-restrained braced frames (BRBFs), and special moment frames (SMFs) for an 8-story building, delivering biweekly professional design reports.
- Calculated demand and selected reinforced concrete beam and beam-column connection sizing per ACI 318 provisions, then produced construction-quality structural drawings in SolidWorks for team design review.
- Determined governing gravity and seismic loads and computed effective seismic weight per ASCE 7 to support lateral design of a special reinforced concrete shear wall system.
- Designed a two-way reinforced concrete slab system and a cantilever retaining wall with combined spread footing, evaluating soil bearing capacity and sliding/overturning stability under lateral earth pressure.

PCI Big Beam Competition **San Diego, CA**
National Precast/Prestressed Concrete Design Competition *Mar 2026 – Jun 2026*

- Collaborated in a 4-student team to design a precast, pretensioned concrete I-beam for a 17-ft simply supported span carrying two 10-kip unfactored service point loads (32-kip total factored load), per ACI 318-19 and the PCI Design Handbook (8th ed.).
- Derived sectional properties and prestress losses to guide initial design iterations, then performed flexural analysis to predict cracking load, ultimate capacity, and midspan deflection against PCI's design-accuracy criteria.
- Modeled and graphed deflections under unfactored, factored, cracking, and peak loading conditions, and selected rebar eccentricity and allowable stresses to prevent top-fiber tension at transfer while minimizing material cost and beam weight.
- Co-authored the structural design section of the team's technical report, documenting calculation methodology and predicted-vs-tested performance across PCI's design accuracy, cost, weight, and deflection judging categories.

C.L.A.S.P. (Classical Lamination Analytical Solver Program) **San Diego, CA**
Independent Software Project *Dec 2025 – Jul 2026*

- Rebuilt a MATLAB-based Classical Laminate Theory (CLT) solver into a standalone, dependency-free browser application, implementing ABD stiffness matrix derivation, ply-by-ply stress/strain recovery, and thermal load/moment (N^T, M^T) equivalence in vanilla JavaScript.
- Implemented Max Stress, Max Strain, and Hashin–Rotem (quadratic) failure criteria with an iterative bisection search to compute first-ply failure loads under independent per-component load scaling, supporting arbitrary ply counts, orientations, and multiple materials per layup.
- Validated numerical accuracy by independently re-deriving the ABD matrix in a separate line-by-line implementation and cross-checking against the original MATLAB output, achieving agreement to 6 significant figures across multiple layups and thermal loading cases.
- Built a Playwright-based automated regression test suite covering UI interactions and numerical correctness, then deployed the tool as a static, offline-capable web app on GitHub Pages supporting both Imperial and Metric unit systems.

SKILLS & INTERESTS

Structural Analysis & Design: Finite Element Analysis, Hand Calculations, Load Path Analysis, Structural Modeling (2D/3D frames, trusses, shells), Material Behavior
Software: AutoCAD, SolidWorks, ABAQUS, MATLAB, Python, Excel
Languages: English (native), Vietnamese (native)